E9 Trigonometric functions in context A level only

Name: ______________________

Class: ______________________

Date: ______________________

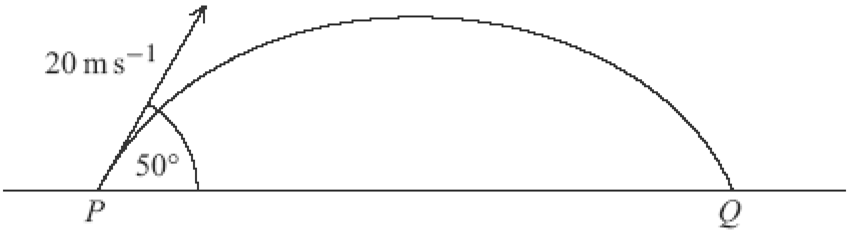
Time: **464 minutes**

Marks: **385 marks**

Comments:

Q1.

A ball is kicked from the point P on a horizontal surface. It leaves the surface with a velocity of 20 m s^{-1} at an angle of 50° above the horizontal and hits the surface for the first time at the point Q. Assume that the ball is a particle that moves only under the influence of gravity.



(a) Show that the time that it takes the ball to travel from P to Q is 3.13 s, correct to three significant figures.

(4)

(b) Find the distance between the points P and Q.

(2)

(c) If a heavier ball were projected from P with the same velocity, how would the distance between P and Q, calculated using the same modelling assumptions, compare with your answer to part (b)? Give a reason for your answer.

(2)

(d) Find the maximum height of the ball above the horizontal surface.

(3)

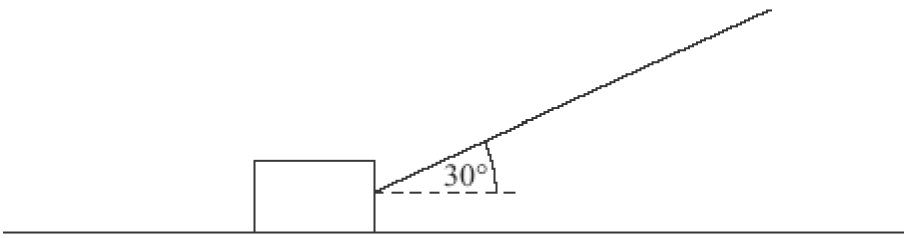
(e) State the magnitude and direction of the velocity of the ball as it hits the surface.

(2)

(Total 13 marks)

Q2.

The diagram shows a block, of mass 20 kg, being pulled along a rough horizontal surface by a rope inclined at an angle of 30° to the horizontal.



The coefficient of friction between the block and the surface is μ. Model the block as a particle which slides on the surface.

(a) If the tension in the rope is 60 newtons, the block moves at a constant speed.

(i) Show that the magnitude of the normal reaction force acting on the block is 166 N.

(3)

(ii) Find μ.

(4)

(b) If the rope remains at the same angle and the block accelerates at 0.8 m s^{-2}, find the tension in the rope.

(5)

(Total 12 marks)

Q3.

A pellet is fired from a window at a height of 3 metres above horizontal ground. Initially, the pellet travels at 70 m s^{-1} at an angle of 10° above the horizontal.

(a) Show that the time for which the pellet travels before it hits the ground is 2.71 seconds, correct to three significant figures.

(6)

(b) Find the horizontal distance that the pellet travels before it hits the ground.

(2)

(c) Find the minimum speed of the pellet during its flight.

(2)

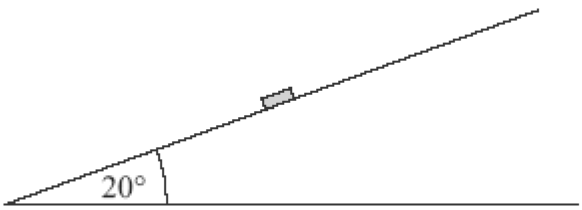
(d) Find the speed of the pellet when it hits the ground.

(4)

(Total 14 marks)

Q4.

A puck, of mass 0.2 kg, is placed on a slope inclined at 20° above the horizontal, as shown in the diagram.



The puck is hit so that initially it moves with a velocity of 4 m s^{-1} directly up the slope.

(a) A simple model assumes that the surface of the slope is smooth.

(i) Show that the acceleration of the puck up the slope is –3.35 m s^{-2}, correct to three significant figures.

(3)

(ii) Find the distance that the puck will travel before it comes to rest.

(3)

(iii) What will happen to the puck after it comes to rest?

Explain why.

(2)

(b) A revised model assumes that the surface is rough and that the coefficient of friction between the puck and the surface is 0.5.

(i) Show that the magnitude of the friction force acting on the puck during this motion is 0.921 N, correct to three significant figures.

(3)

(ii) Find the acceleration of the puck up the slope.

(3)

(iii) What will happen to the puck after it comes to rest in this case?

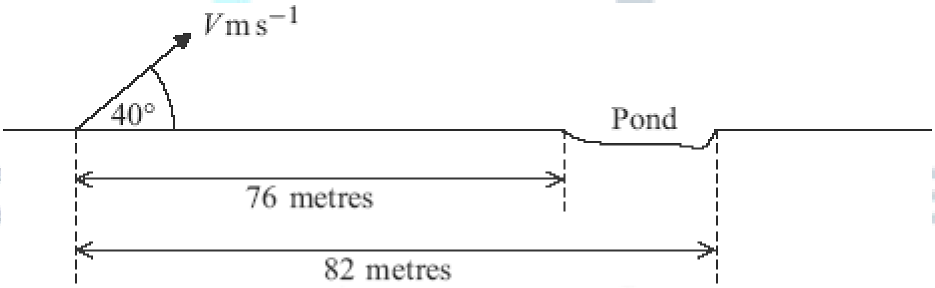
Explain why.

(2)

(Total 16 marks)

Q5.

A golfer hits a ball which is on horizontal ground. The ball initially moves with speed V m s^{-1} at an angle of 40° above the horizontal. There is a pond further along the horizontal ground. The diagram below shows the initial position of the ball and the position of the pond.



(a) State **two** assumptions that you should make in order to model the motion of the ball.

(2)

(b) Show that the horizontal distance, in metres, travelled by the ball when it returns to ground level is

$$\frac{V^2 \sin 40° \cos 40°}{4.9}$$

(6)

(c) Find the range of values of V for which the ball lands in the pond.

(4)

(Total 12 marks)

Q6.

A ball is kicked so that it leaves a horizontal surface, at the point , travelling at 16 m s^{-1} and at an angle θ above the horizontal. The ball hits the surface again 2 seconds later, at the point B. Assume that the ball is a particle that moves only under the influence of gravity.

(a) Show that θ = 37.8°, correct to three significant figures.

(3)

(b) Find the time for which the ball is more than 2 metres above the surface.

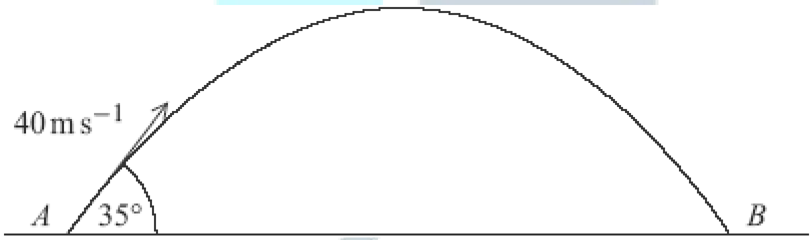
(5)

(Total 8 marks)

Q7.

A ball is hit by a bat so that, when it leaves the bat, its velocity is 40 m s^{-1} at an angle of 35° above the horizontal. Assume that the ball is a particle and that its weight is the only force that acts on the ball after it has left the bat.

(a) A simple model assumes that the ball is hit from the point A and lands for the first time at the point B, which is at the same level as A, as shown in the diagram.



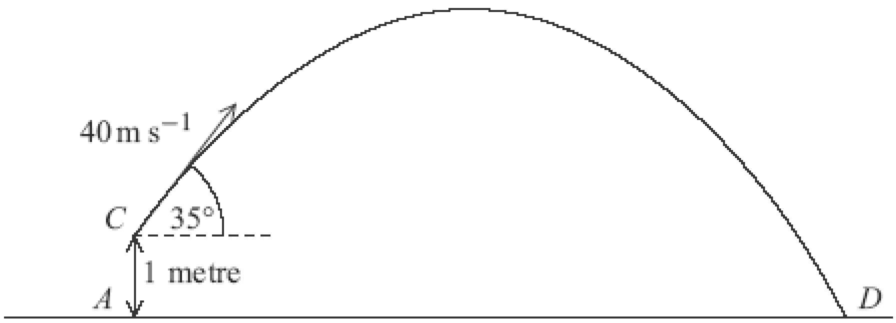
(i) Show that the time that it takes for the ball to travel from A to B is 4.68 seconds, correct to three significant figures.

(4)

(ii) Find the horizontal distance from A to B.

(2)

(b) A revised model assumes that the ball is hit from the point C, which is 1 metre above A. The ball lands at the point D, which is at the same level as A, as shown in the diagram.



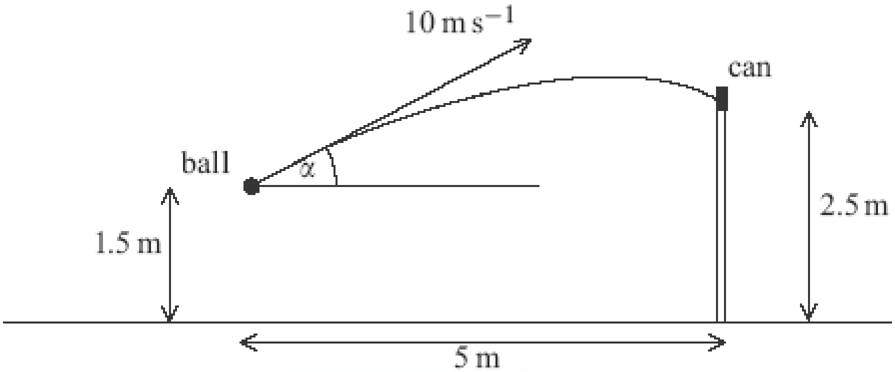
Find the time that it takes for the ball to travel from C to D.

(6)

(Total 12 marks)

Q8.

A boy throws a small ball from a height of 1.5 m above horizontal ground with initial velocity 10 m s^{-1} at an angle α above the horizontal. The ball hits a small can placed on a vertical wall of height 2.5 m, which is at a horizontal distance of 5 m from the initial position of the ball, as shown in the diagram.



(a) Show that α satisfies the equation

$$49 \tan^2 \alpha - 200 \tan \alpha + 89 = 0$$

(7)

(b) Find the two possible values of α, giving your answers to the nearest 0.1°.

(3)

(c) (i) To knock the can off the wall, the horizontal component of the velocity of the ball must be greater than 8 m s^{-1}.

Show that, for one of the possible values of α found in part (b), the can will be knocked off the wall, and for the other, it will **not** be knocked off the wall.

(3)

(ii) Given that the can is knocked off the wall, find the direction in which the ball is moving as it hits the can.

(4)

(Total 17 marks)

Q9.

A cricket ball is hit from the floor of a sports hall, which has a height of 6 metres. The initial velocity of the ball is 20 m s^{-1} at an angle of 60° above the horizontal.

Assume that the cricket ball is a particle which moves only under the influence of gravity.

(a) Show that the ball hits the ceiling of the sports hall approximately 0.389 seconds after it was hit.

(5)

(b) Find the horizontal distance travelled by the ball before it hits the ceiling.

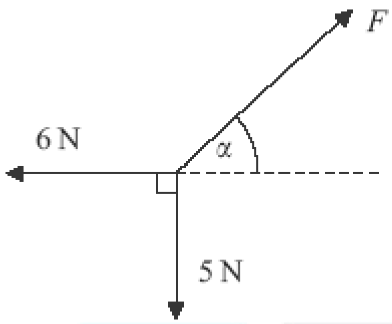
(2)

(c) Find the speed of the ball just before it hits the ceiling.

(5)
(Total 12 marks)

Q10.

The diagram shows three forces which act in the same plane and are in equilibrium.



(a) Find F.

(3)

(b) Find α.

(3)
(Total 6 marks)

Q11.

A trolley, of mass 100 kg, rolls at a constant speed along a straight line down a slope inclined at an angle of 4° to the horizontal.

Assume that a constant resistance force, of magnitude P newtons, acts on the trolley as it moves. Model the trolley as a particle.

(a) Draw a diagram to show the forces acting on the trolley.

(1)

(b) Show that P = 68.4 N, correct to three significant figures.

(3)

(c) (i) Find the acceleration of the trolley if it rolls down a slope inclined at 5° to the horizontal and experiences the same constant force of magnitude P that you found in part (b).

(4)

(ii) Make one criticism of the assumption that the resistance force on the trolley is constant.

(1)

(Total 9 marks)

Q12.

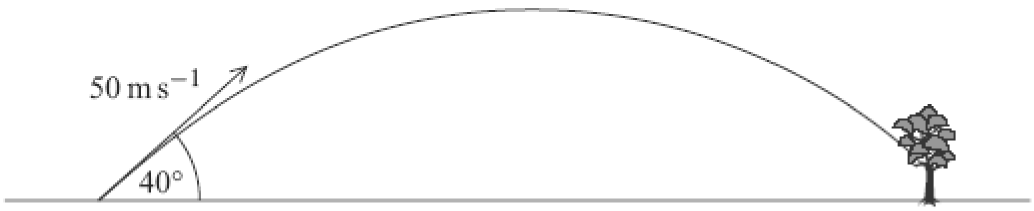
A golf ball is struck from a point on horizontal ground so that it has an initial velocity of 50 m s^{-1} at an angle of 40° above the horizontal.

Assume that the golf ball is a particle and its weight is the only force that acts on it once it is moving.

(a) Find the maximum height of the golf ball.

(4)

(b) After it has reached its maximum height, the golf ball descends but hits a tree at a point which is at a height of 6 metres above ground level.

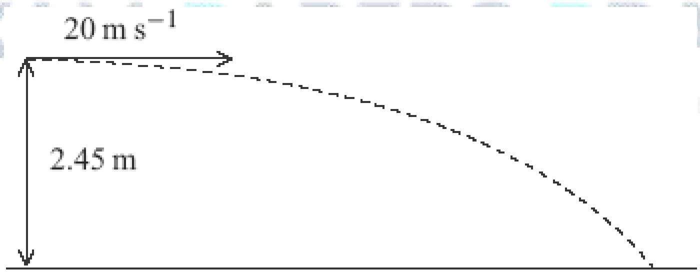


Find the time that it takes for the ball to travel from the point where it was struck to the tree.

(6)
(Total 10 marks)

Q13.

A tennis ball is hit from a height of 2.45 metres above horizontal ground. Initially it travels horizontally at a speed of 20 m s^{-1}, as shown in the diagram.



(a) Show that the time taken for the tennis ball to reach the ground is 0.707 seconds, correct to three significant figures.

(3)

(b) Find the horizontal distance travelled by the ball when it hits the ground.

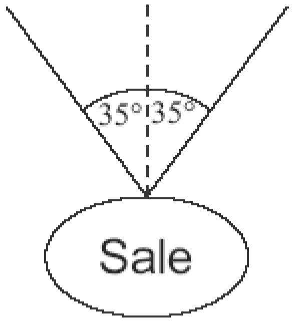
(2)

(c) Find the angle between the velocity of the ball and the horizontal when the ball hits the ground.

(4)
(Total 9 marks)

Q14.

A sign, of mass 2 kg, is suspended from the ceiling of a supermarket by two light strings. It hangs in equilibrium with each string making an angle of 35° to the vertical, as shown in the diagram. Model the sign as a particle.



(a) By resolving forces horizontally, show that the tension is the same in each string.

(2)

(b) Find the tension in each string.

(5)

(c) If the tension in a string exceeds 40 N, the string will break. Find the mass of the heaviest sign that could be suspended as shown in the diagram.

(3)

(Total 10 marks)

Q15.

A box, of mass 3 kg, is placed on a slope inclined at an angle of 30° to the horizontal. The box slides down the slope. Assume that air resistance can be ignored.

(a) A simple model assumes that the slope is smooth.

(i) Draw a diagram to show the forces acting on the box.

(1)

(ii) Show that the acceleration of the box is 4.9 m s^{-2}.

(2)

(b) A revised model assumes that the slope is rough. The box slides down the slope from rest, travelling 5 metres in 2 seconds.

(i) Show that the acceleration of the box is 2.5 m s^{-2}.

(2)

(ii) Find the magnitude of the friction force acting on the box.

(3)

(iii) Find the coefficient of friction between the box and the slope.

(5)

(iv) In reality, air resistance affects the motion of the box. Explain how its acceleration would change if you took this into account.

(2)
(Total 15 marks)

Q16.

An arrow is fired from a point A with a velocity of 25 m s^{-1}, at an angle of 40° above the horizontal. The arrow hits a target at the point B which is at the same level as the point A, as shown in the diagram.

(a) State **two** assumptions that you should make in order to model the motion of the arrow.

(2)

(b) Show that the time that it takes for the arrow to travel from A to B is 3.28 seconds, correct to three significant figures.

(4)

(c) Find the distance between the points A and B.

(2)

(d) State the magnitude and direction of the velocity of the arrow when it hits the target.

(2)

(e) Find the minimum speed of the arrow during its flight.

(2)
(Total 12 marks)

Q17.

A ball is projected with speed u m s^{-1} at an angle of elevation a above the horizontal so as to hit a point P on a wall. The ball travels in a vertical plane through the point of projection. During the motion, the horizontal and upward vertical displacements of the ball from the point of projection are x metres and y metres respectively.

(a) Show that, during the flight, the equation of the trajectory of the ball is given by

$$y = x\tan\alpha - \frac{gx^2}{2u^2}\left(1+\tan^2\alpha\right)$$

(6)

(b) The ball is projected from a point 1 metre vertically below and R metres horizontally from the point P.

(i) By taking g = 10 m s^{-2}, show that R satisfies the equation

$$5R^2 \tan^2 \alpha - u^2 R \tan \alpha + 5R^2 + u^2 = 0$$

(2)

(ii) Hence, given that u and R are constants, show that, for tan a to have real values, R must satisfy the inequality

$$R^2 \leq \frac{u^2(u^2 - 20)}{100}$$

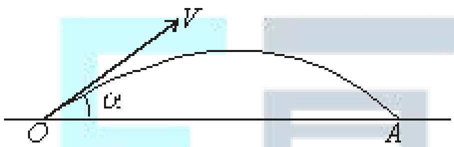
(2)

(iii) Given that R = 5, determine the minimum possible speed of projection.

(3)
(Total 13 marks)

Q18.

A golf ball is projected from a point O with initial velocity V at an angle α to the horizontal. The ball first hits the ground at a point which is at the same horizontal level as O, as shown in the diagram.



It is given that V cos α = 6u and V sin α = 2.5u.

(a) Show that the time taken for the ball to travel from O to A is $\frac{5u}{g}$.

(4)

(b) Find, in terms of g and u, the distance OA.

(2)

(c) Find V in terms of u.

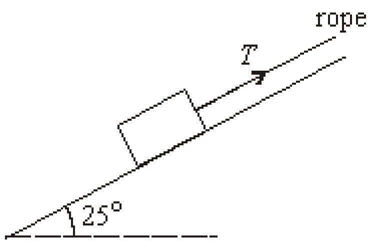
(2)

(d) State, in terms of u, the least speed of the ball during its flight from O to A.

(1)
(Total 9 marks)

Q19.

A rough slope is inclined at an angle of 25° to the horizontal. A box of weight 80 newtons is on the slope. A rope is attached to the box and is parallel to the slope. The tension in the rope is of magnitude T newtons. The diagram shows the slope, the box and the rope.



(a) The box is held in equilibrium by the rope.

(i) Show that the normal reaction force between the box and the slope is 72.5 newtons, correct to three significant figures.

(3)

(ii) The coefficient of friction between the box and the slope is 0.32. Find the magnitude of the maximum value of the frictional force which can act on the box.

(2)

(iii) Find the least possible tension in the rope to prevent the box from moving down the slope.

(4)

(iv) Find the greatest possible tension in the rope.

(3)

(v) Show that the mass of the box is approximately 8.16 kg.

(1)

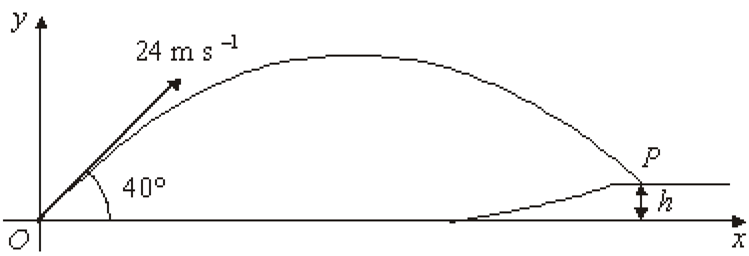
(b) The rope is now released and the box slides down the slope. Find the acceleration of the box.

(3)

(Total 16 marks)

Q20.

A golf ball is struck from a point O with velocity 24 m s^{-1} at an angle of 40° to the horizontal. The ball first hits the ground at a point P, which is at a height h metres above the level of O.



The horizontal distance between O and P is 57 metres.

(a) Show that the time that the ball takes to travel from O to P is 3.10 seconds, correct to three significant figures.

(3)

(b) Find the value of h.

(3)

(c) (i) Find the speed with which the ball hits the ground at P.

(5)

(ii) Find the angle between the direction of motion and the horizontal as the ball hits the ground at P.

(2)

(Total 13 marks)

Q21.

A football is kicked from a point O on a horizontal football ground with a velocity of 20 m s^{-1} at an angle of elevation of 30°. During the motion, the horizontal and upward vertical displacements of the football from O are x metres and y metres respectively.

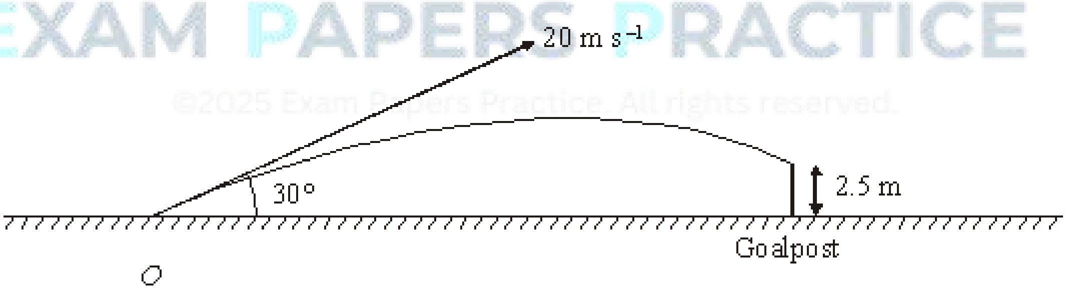
(a) Show that x and y satisfy the equation

$$y = x \tan 30° - \frac{gx^2}{800\cos^2 30°}$$

(6)

(b) On its downward flight the ball hits the horizontal crossbar of the goal at a point which is 2.5 m above the ground. Using the equation given in part (a), find the horizontal distance from O to the goal.

(4)



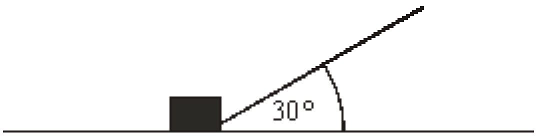
(c) State **two** modelling assumptions that you have made.

(2)

(Total 12 marks)

Q22.

The diagram shows a rope that is attached to a box of mass 25 kg, which is being pulled along rough horizontal ground. The rope is at an angle of 30° to the ground. The tension in the rope is 40 N. The box accelerates at 0.1 m s^{-2}.



(a) Draw a diagram to show all of the forces acting on the box.

(1)

(b) Show that the magnitude of the friction force acting on the box is 32.1 N, correct to three significant figures.

(3)

(c) Show that the magnitude of the normal reaction force that the ground exerts on the box is 225 N.

(3)

(d) Find the coefficient of friction between the box and the ground.

(2)

(Total 9 marks)

Q23.

A golf ball is struck so that its initial velocity is 30 m s^{-1} at an angle of 60° above the horizontal. Assume that the ground is horizontal.

(a) (i) Show that the time of flight of the particle is 5.30 seconds, correct to three significant figures.

(3)

(ii) Find the distance of the ball from its initial position when it hits the ground for the first time.

(2)

(b) In fact, the ball hits a tree at a point which is at a height of 5 metres above the ground. Given that the ball is descending when it hits the tree, calculate the distance of the tree from the point where the ball was struck.

(6)

(Total 11 marks)

Q24.

A football is placed on a horizontal surface. It is then kicked, so that it has an initial velocity of 12 m s^{-1} at an angle of 40° above the horizontal.

(a) State two modelling assumptions that it would be appropriate to make when considering the motion of the football.

(2)

(b) (i) Find the time that it takes for the ball to reach its maximum height.

(4)

(ii) Hence show that the maximum height of the ball is 3.04 metres, correct to three significant figures.

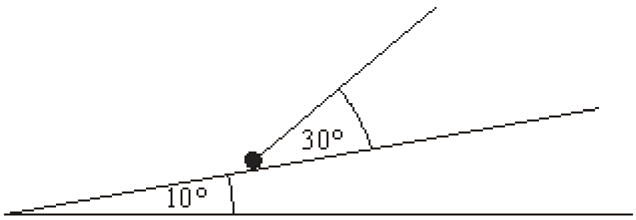
(3)

(c) After the ball has reached its maximum height, it hits the bar of a goal at a height of 2.44 metres. Find the horizontal distance of the goal from the point where the ball was kicked.

(7)
(Total 16 marks)

Q25.

A rough slope is inclined at an angle of 10° to the horizontal. A particle of mass 6 kg is on the slope. A string is attached to the particle and is at an angle of 30° to the slope. The tension in the string is 20 N. The diagram shows the slope, the particle and the string.



The particle moves up the slope with an acceleration of 0.4 m s^{-2}.

(a) Draw a diagram to show the forces acting on the particle.

(1)

(b) Show that the magnitude of the normal reaction force is 47.9 N, correct to three significant figures.

(4)

(c) Find the coefficient of friction between the particle and the slope.

(6)
(Total 11 marks)

Q26.

A golf ball is driven with speed 40 m s^{-1} at an angle α to the horizontal from a point on a horizontal golf course. The ball travels in a vertical plane.

(a) Show that, during its flight, the horizontal and vertical displacements, x metres and y metres respectively, of the ball from the point of projection satisfy the equation

$$y = x \tan \alpha - \frac{gx^2}{3200}(1 + \tan^2 \alpha)$$

(5)

(b) The golf ball just clears a tree of vertical height 4 m at a horizontal distance of 100 m from its point of projection.

Find the two possible values of α.

(6)

(c) State two modelling assumptions that you have made.

(2)
(Total 13 marks)

Q27.

A car moves in a straight line up a hill which is inclined at an angle of α to the horizontal, where $\sin \alpha = \frac{1}{14}$. The mass of the car is 1200 kg.

(a) During the first part of its motion, the car moves with constant speed and is subject to a propulsive force of 1500 newtons and a resistance force of R newtons.

(i) Draw a diagram showing the forces acting on the car.

(2)

(ii) Show that $R = 660$.

(4)

(iii) The car travels 60 metres in 4 seconds. Write down the speed of the car.

(1)

(b) During the second part of its motion, the propulsive force ceases to act and the resistance force is still 660 newtons. The car continues to move up the hill until it comes to rest.

(i) Find the magnitude of the retardation of the car in this stage.

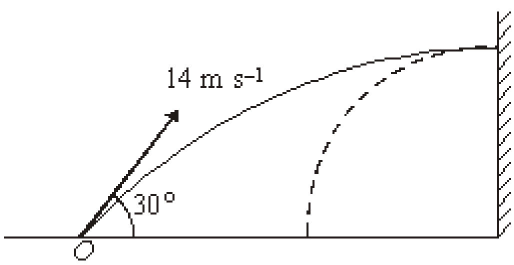
(2)

(ii) Find the time between the propulsive force ceasing to act and the car coming to rest.

(2)
(Total 11 marks)

Q28.

Adam kicks his ball from a point O on a horizontal surface towards a vertical wall. He kicks the ball with speed 14 m s^{-1} at an angle of 30° to the horizontal, as shown in the diagram.



When the ball hits the wall, it is travelling **horizontally**.

(a) Find the vertical height above O at which the ball hits the wall.

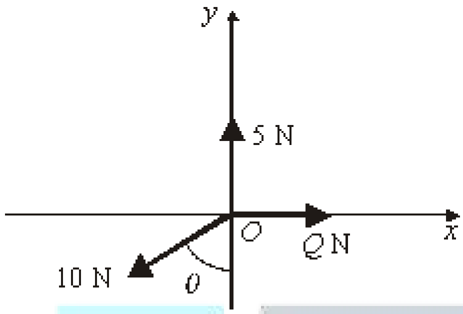
(4)

(b) The ball rebounds horizontally from the wall. Find the time that the ball takes to reach the horizontal surface.

(3)
(Total 7 marks)

Q29.

A particle rests in equilibrium at a point O on a horizontal surface. It is acted on by three horizontal forces of magnitudes 5 N, 10 N and Q N. The directions of these forces, relative to the horizontal axes Ox and Oy, are shown in the diagram below.



(a) Show that θ = 60°.

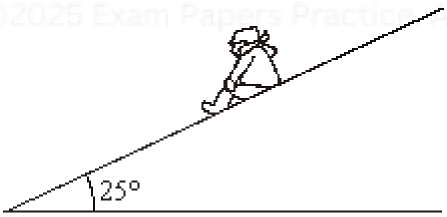
(2)

(b) Find the value of Q.

(2)
(Total 4 marks)

Q30.

A children's slide is straight and inclined at 25° to the horizontal, as shown in the diagram.



Matthew, of mass 35 kg, goes down the slide at constant speed.

(a) Draw a diagram to show the forces acting on Matthew.

(1)

(b) Find the magnitude of the normal reaction force between Matthew and the slide.

(3)

(c) Find the coefficient of friction between Matthew and the slide.

(4)

(Total 8 marks)

Q31.

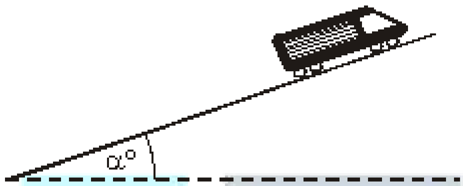
A mountain railway train moves on a straight track. The mass of the train and its passengers is 1000 kg. During its motion the train moves under the action of a variable propulsive force, P newtons, and a constant resistance force of R newtons.

(a) During the first stage of its motion, the train moves horizontally with acceleration 0.25 m s^{-2}. In this stage, the value of P is 1200.

Show that R = 950.

(3)

(b) During the second stage of its motion, the train moves up a slope inclined at an angle α to the horizontal, where sin α = 0.1.



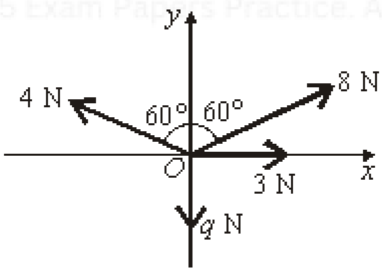
In this stage, the value of P = 2100, and the value of R remains at 950. Find the acceleration of the train.

(4)

(Total 7 marks)

Q32.

A particle is at a point O on a smooth horizontal surface. It is acted on by four horizontal forces of magnitudes 3 N, 8 N, 4 N and q N. The directions of these forces, relative to the horizontal axes Ox and Oy, are shown in the diagram below.



The resultant, **R** newtons, of these forces acts along the line Ox.

(a) Show that q = 6.

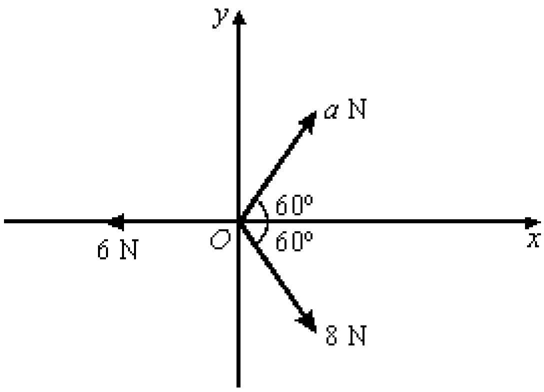
(4)

(b) Find the magnitude of **R**.

(3)

(Total 7 marks)

Q33.



A particle is at a point O on a smooth horizontal surface. It is acted on by three horizontal forces of magnitudes 6 N, 8 N and N. Relative to horizontal axes Ox and Oy, the directions of these three forces are shown in the diagram. The resultant, **R**, of these forces acts along the line Oy.

(a) Show that a = 4.

(4)

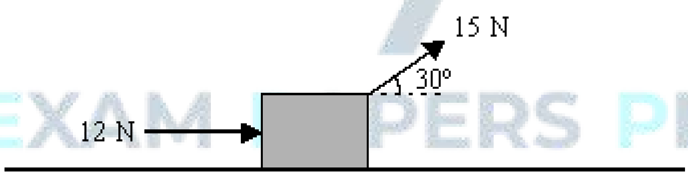
(b) Find the magnitude of **R**.

(3)

(Total 7 marks)

Q34.

The diagram shows a small box resting on a rough horizontal surface.



The box is of weight W newtons. It is pushed with a horizontal force of 12 newtons, and pulled with a force of 15 newtons at an angle of 30° to the horizontal.

The box rests in limiting equilibrium.

(a) Draw a diagram to show all the forces acting on the box.

(2)

(b) Show that the frictional force acting on the box is approximately 25 newtons.

(3)

(c) The coefficient of friction between the box and the surface is $\frac{1}{3}$
Find the normal reaction force between the box and the surface.

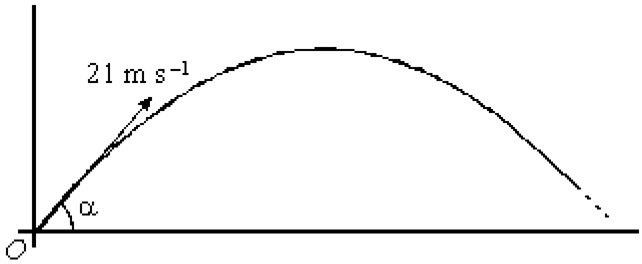
(2)

(d) Find the value of W.

(3)
(Total 10 marks)

Q35.

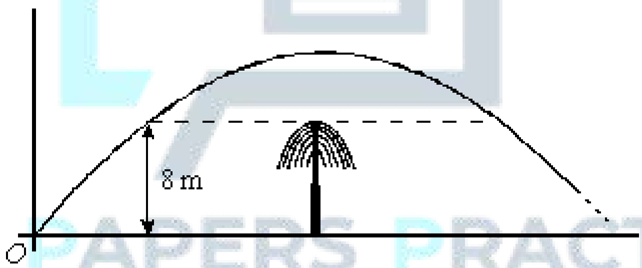
Paul throws a ball from point O with a velocity 21 m s^{-1} at an angle of α to the horizontal, where sin α = 0.7. The ball subsequently moves freely under gravity in a vertical plane, as shown in the diagram.



(a) Show that the time taken for the ball to reach its greatest height above O is 1.5 seconds.

(3)

(b) When the ball reaches its greatest height, it passes over a tree of vertical height 8 metres, as shown in the diagram below.



(i) Find the vertical distance between the ball and the top of the tree at this time.

(4)

(ii) Find the time between the ball leaving O and first reaching the horizontal level of the top of the tree. Give your answer to two decimal places.

(5)

(iii) Find the length of time for which the ball is above the horizontal level of the top of the tree.

(2)
(Total 14 marks)